

L ASHWINI BALACHANDRAN

+91 9667712326 New Delhi, India lashwinib@gmail.com

linkedin.com/in/ashwini-lb chickenleaf.wordpress.com/ github.com/chickenleaf leetcode.com/u/chickenleaf

EDUCATION

Dual Degree(MS Management, BTech Computer Science)

2024

Jawaharlal Nehru University, New Delhi

First Division

EXPERIENCE

SDE Intern (Rust & Python)

Dec 2024 - Mar 2025

[The Linux Foundation](#) & The Mozilla Foundation

Remote

- **Designed and implemented** a new **platform-agnostic dialog subsystem** in the **Servo browser engine** using enums, with full ownership of state management, event dispatch, GUI rendering (egui), and engine integration.
- **Extended and modified** the **Embedder API** and used **Serde** for serializing dialog messages across the IPC boundary to **safely propagate** dialog state and user interactions between the GUI layer and browser engine, ensuring **consistent and isolated dialog behavior** across webview instances.
- **Implemented** the **DOMRectList interface** for the Servo engine, adhering to W3C specifications and enabling support for legacy web content like `getClientRects()`. Ensured correct indexed access, length handling, and memory safety with Rust's type system and custom GC-safe tracing.
- **Refactored** `Element.getClientRects()` to **return** a standards-compliant **DOMRectList**, improving layout box calculations and transforming data across HTML and SVG elements.
- **Refactored garbage collection (GC) handling** in Servo's Rust codebase by replacing implicit `CanGc::note()` calls with explicit `CanGc` arguments across **2,600+ lines of code**. This made **GC hazards more visible**, **improved memory safety**, and **clarified ownership in backend systems**.
- Applied context-specific rules: added `CanGc` in bare impl methods, preserved `CanGc::note()` in `task!` closures and external traits, **updated Bindings.conf** for **FooMethods WebIDL traits** to include `canGc` property for **correct WebIDL codegen**. Also created **wrapper functions** to bridge updated signatures with legacy interfaces, ensuring **FFI and API compatibility**.
- **Addressed several issues around memory safety, code clarity, concurrency, and performance in Servo's codebase**. Fixed problems like unsafe transmute, late/zero-filled initialization, and refactored long complex types using type aliases and structs. Boxed large enum variants and ensured thread safety with smart pointers and wrapper structs for safe `Send/Sync` usage.

LLM Engineer Intern

Feb 2024 - Jul 2024

[Sekurenet](#)

Remote

- Reduced query processing time by 15% using **Redis** caching and **FAISS** while building **Q&A agents** and **RAG bots** with open-source **LLMs** (Mistral, Mixtral, LiteLLM, Llama), **Qdrant**, **sentence transformers**, and **Hugging Face NLP** libraries.
- Built & integrated **AI assistants** for email and calendar tools using **APIs** and python libraries, improving user productivity by 10% with **Phidata**, **VapiAI**, **LangChain**, **OpenAI**, **Ollama**, and **Groq Client**.
- Developed ClickUp **RESTful API** tools and **PythonREPL** script, enhancing testing efficiency for the **team**.
- Explored **self-operating computers** with **LLMs**, **YOLO**, **LLaVA**, **Moondream**, **SAM**, and **PyAutoGUI**.

SDE Intern (Python)

Jan 2023 - Jun 2023

[ThirdOrigin](#)

Remote

- Reduced processing time by 85.7% with **asynchronous programming** and **batch loading** for **MongoDB**.
- Cut data loading time from 14 minutes to under 2 minutes using **Celery**, **Redis**, and **multithreading** in batch sizes of 10,000+ optimized via **CPU**, **time** and **memory profiling**.
- Developed **secure CRUD** operations for **RDBMS** (MySQL, PostgreSQL) and **RESTful APIs** for **NoSQL** time-series databases (**QuestDB**, **InfluxDB**), using **Pydantic** & **ORM** to prevent **SQL injection** and reduce input errors by 40%.
- Optimized various libraries by atleast 20% using **profiling**, **batch processing**, **data structures**, and **testing**.

SKILLS

Languages	Rust, Python, C/C++, JavaScript
Databases	RDBMS (PostgreSQL, MySQL), NoSQL (MongoDB, Redis, InfluxDB, QuestDB), RabbitMQ
Tools	Django, Flask, FastAPI, Docker, Kubernetes, Celery, Git, Postman, CI/CD, Nginx, ElasticStack, MERN
ML/AI	scikit-learn, GridSearchCV, Hugging Face Transformers, LangChain, OpenCV, YOLO, SAM

PROJECTS

Web Portfolio: Django, SQLite, Render, Docker, CI/CD, Webhook, Gunicorn, Whitenoise, Networking.

QueryJet API: FastAPI, Elasticsearch, Celery, RabbitMQ, Docker, Kubernetes.

Kaggle Titanic- ML Notebook: Pandas, Matplotlib, scikit-learn, Support Vector Classifier, Random Forest, GridSearchCV.

NativeFlow: Transformers Library, MarianMT, Langdetect, NLP, FastAPI, SQLAlchemy, PostgreSQL, Docker.

ACHIEVEMENTS

- Selected as **1 of 26** interns **worldwide** for a highly competitive **open source** internship program (Outreachy'24 Dec cohort).
- Ranked in the **top 1%** in the Kaggle Titanic Machine Learning competition leaderboard.
- Solved more than **220+** **DSA** problems on **Leetcode** with **consistency**, earning multiple **monthly** and **annual** badges.